ARDEN CLIMATE VULNERABILITY ASSESSMENT

Third edition, prepared for the Environmental Quality Board

1. SCOPE

This assessment describes exposure and sensitivity across the State under a range of scenarios. It is a planning document. It does not value loss and is not a basis for any financial estimate.

2. COASTAL EXPOSURE

The State has 313 miles of tidal shoreline including the offshore islands. Mean sea level at the Ardenport gauge has risen 1.0 feet over the record. Under the intermediate scenario the assessment projects an additional 1.2 feet by mid-century, which would place 20,362 additional parcels within the tidal flood plain.

3. INLAND FLOODING

The assessment identifies 11 catchments in which the frequency of the design storm has increased materially over the record. The Millvale catchment ranks first on the assessment's composite index.

4. HEAT AND DROUGHT

Days above ninety degrees have increased from a mean of 16 to 27 over the record. The assessment notes consequences for the agricultural counties and for electricity demand but treats them as outside the scope of the coastal and flood chapters.

5. SOCIAL VULNERABILITY

The assessment applies a composite index of social vulnerability. 30 census tracts fall in the highest quintile on both exposure and vulnerability; 7 of them are in the Millvale region.

6. LIMITATIONS

Scenario projections are not predictions. The assessment does not attribute any individual event to any cause and does not apportion responsibility for any loss.

Prepared by the Board's science advisory panel. The Board has neither adopted nor rejected the panel's scenarios.